Notes on the Completion of the Space of Continuous Functions under Integral Norms

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January 8, 2024

1 A Brief Summary of Riemann-Stieltjes Integral

In the context of probability, if a random variable has a probability density function $p(x)$, then the probability that the variable falls into $[a, b]$ is given by $\int_a^b p(x)\,dx$. Such a random variable can't have a positive probability taking on a single value. A general random variable X has a cumulative distribution function $F(x) := \mathcal{P}(x : X \leq x)$. It is a non-decreasing function of x---we will also use the terminology increasing function interchangeably. The Riemann-Stieltjes integral with respect to an increasing function is a step in describing the probability of a general random variable in terms of integrals.

In the context of functional analysis, there is a need to consider possible convergence of a sequence of linear functional on $C[a, b]$, given in the form of $g \in C[a, b] \mapsto \int_a^b g(x) f_k(x)\,dx$, where $\{f_k\}$ is a sequence of continuous or Riemann integrable functions on $[a, b]$ with certain bounds, say, $\int_a^b |f_k(x)|\,dx \leq M$ for all k. If there is a limit, then it could be given in terms of the Riemann-Stieltjes integral with respect to some increasing function.

Given a monotone increasing function α on $[a, b]$, we can mimic the steps in defining the usual Riemann integral of a given bounded function f on $[a, b]$ to define the lower sum

$$L(f, \mathcal{P}, d\alpha) := \sum_{i=1}^{k} m_{I_i}(f) d\alpha(I_i)$$

and upper sum

$$U(f, \mathcal{P}, d\alpha) := \sum_{i=1}^{k} M_{I_i}(f) d\alpha(I_i)$$

of f with respect to a partition $\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_k = b\}$ of $[a, b]$, with $I_i = [x_{i-1}, x_i]$, $d\alpha(I_i) = \alpha(x_i) - \alpha(x_{i-1})$, $m_{I_i}(f) = \inf_{I_i} f$, and $M_{I_i}(f) = \sum_{I_i} f$.

Definition 1.1 Suppose that f is a bounded function defined on the interval $[a, b]$. Then its upper integral, and respectively lower integral, on $[a, b]$ with respect to α is defined as $\inf_{\mathcal{P}} U(f, \mathcal{P}, \alpha)$, and respectively $\sup_{\mathcal{P}} U(f, \mathcal{P}, \alpha)$, where $\mathcal{P}$ runs over all partitions of $\mathcal{P}$.

The upper integral is denoted as $\overline{\int_a^b} f\,d\alpha$, while the lower integral is denoted as $\underline{\int_a^b} f\,d\alpha$. $\diamondsuit$

When $\alpha(x) = x$, we write $U(f, \mathcal{P})$ for $U(f, \mathcal{P}, dx)$, and $L(f, \mathcal{P})$ for $L(f, \mathcal{P}, dx)$.

Exercise 1.2 Let $\mathcal{C}$ denote the standard tertiary Cantor set on $[0, 1]$ and $\chi_{\mathcal{C}}$ denote its characteristic function which takes value 1 on $\mathcal{C}$ and 0 elsewhere. Let $\mathcal{P}$ be a partition of $[0, 1]$ into intervals of equal length 3^{-k} for some $k \in \mathbb{N}$. Find $U(\chi_{\mathcal{C}}, \mathcal{P})$ and $L(\chi_{\mathcal{C}}, \mathcal{P})$. Is there a positive lower bound of $L(\chi_{\mathcal{C}}, \mathcal{P})$ independent of $\mathcal{P}$?

Proposition 1.3 Basic property of upper and lower integral of a bounded real-valued function. *Suppose that f is a bounded function defined on the interval $[a, b]$. Then*

$$\underline{\int_a^b} f\, d\alpha \leq \overline{\int_a^b} f\, d\alpha.$$

Proof. Let $\mathcal{P}_1, \mathcal{P}_2$ be two arbitrary partitions of $[a, b]$, and $\mathcal{P}^*$ be a refinement of both $\mathcal{P}_1$ and $\mathcal{P}_2$. Then

$$L(f, \mathcal{P}_1, d\alpha) \leq L(f, \mathcal{P}^*, d\alpha) \leq U(f, \mathcal{P}^*, d\alpha) \leq U(f, \mathcal{P}_2, d\alpha).$$

As a result,

$$L(f, \mathcal{P}_1, d\alpha) \leq \overline{\int_a^b} f\, d\alpha) = \inf_{\mathcal{P}_2} U(f, \mathcal{P}_2, d\alpha),$$

and

$$\underline{\int_a^b} f\, d\alpha = \sup_{\mathcal{P}_1} L(f, \mathcal{P}_1, d\alpha) \leq \overline{\int_a^b} f\, d\alpha.$$

■

Definition 1.4 A bounded real-valued function f defined on an interval $[a, b]$ is called **Riemann-Stieltjes integrable** with respect to $d\alpha$, if

$$\underline{\int_a^b} f\, d\alpha = \overline{\int_a^b} f\, d\alpha.$$

In such a case we write $f \in \mathcal{R}(\alpha)$ on $[a, b]$, and use $\int_a^b f\, d\alpha$ for $\underline{\int_a^b} f\, d\alpha$. ◊

Theorem 1.5 Riemann-Stieltjes integrability criterion. *A bounded real-valued function f defined on the interval $[a, b]$ is Riemann-Stieltjes integrable iff for any $\epsilon > 0$, there exists a partition $\mathcal{P}$ of $[a, b]$ such that*

$$U(f, \mathcal{P}, d\alpha) - L(f, \mathcal{P}, d\alpha) < \epsilon. \tag{1.1}$$

Exercise 1.6 Define

$$H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases} \text{ and } G(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Determine $\underline{\int_{-1}^1} H(x)\, dH(x)$, $\overline{\int_{-1}^1} H(x)\, dH(x)$, $\underline{\int_{-1}^1} G(x)\, dH(x)$, $\overline{\int_{-1}^1} G(x)\, dH(x)$, and find out if either H or G is in $\mathcal{R}(dH)$ over $[-1, 1]$.

Theorem 1.7 *If f is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.*

Theorem 1.8 *If f is monotone on $[a, b]$ and α continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.*

Theorem 1.9 *If f is bounded on $[a, b]$ and has only finitely many points of discontinuity, and α is continuous at every point at which f is discontinuous, then $f \in \mathcal{R}(\alpha)$ on $[a, b]$.*

Theorem 1.10 Properties of Riemann-Stieltjes Integral. *Suppose that* $f_1, f_2 \in \mathcal{R}(\alpha)$ *on* $[a, b]$.

(a) $f_1 + f_2 \in \mathcal{R}(\alpha)$ *and* $cf_1 \in \mathcal{R}(\alpha)$ *on* $[a, b]$ *for any scalar* c, *and*

$$\int_a^b (f_1 + f_2)\, d\alpha = \int_a^b f_1 d\alpha + \int_a^b f_2\, d\alpha$$
$$\int_a^b cf_1 d\alpha = c \int_a^b f_1 d\alpha.$$

(b) $f_1 f_2 \in \mathcal{R}(\alpha)$.

(c) $|f_1| \in \mathcal{R}(\alpha)$ *and* $\left|\int_a^b f_1 d\alpha\right| \leq \int_a^b |f_1| d\alpha$.

(d) If $f_1(x) \leq f_2(x)$ *on* $[a, b]$, *then*

$$\int_a^b f_1 d\alpha \leq \int_a^b f_2 d\alpha.$$

(e) For any $c, a < c < b$, *then* $f_1 \in \mathcal{R}(\alpha)$ *on* $[a, c]$ *and on* $[c, b]$, *and*

$$\int_a^c f_1 d\alpha + \int_c^b f_1 d\alpha = \int_a^b f_1 d\alpha.$$

(f) If $|f_1(x)| \leq M$ *on* $[a, b]$, *then*

$$\left|\int_a^b f_1 d\alpha\right| \leq M[\alpha(b) - \alpha(b)].$$

(g) If β *is another monotone increasing function on* $[a, b]$ *and* $f_1 \in \mathcal{R}(\beta)$ *on* $[a, b]$, *then* $f_1 \in cR(\alpha + \beta)$ *on* $[a, b]$ *and*

$$\int_a^b f_1 d(\alpha + \beta) = \int_a^b f_1 d\alpha + \int_a^b f_1 d\beta.$$

Theorem 1.11 *Assume that* f *is bounded on* $[a, b]$ *and that* $\alpha' \in \mathcal{R}$ *on* $[a, b]$ *(namely,* α' *is Riemann integrable on* $[a, b]$*). Then* $f \in \mathcal{R}(\alpha)$ *on* $[a, b]$ *iff* $f\alpha' \in \mathcal{R}$ *on* $[a, b]$, *and*

$$\int_a^b f\, d\alpha = \int_a^b f\alpha'\, dx. \tag{1.2}$$

Corollary 1.12 *Assume that* $\alpha' \in \mathcal{R}$ *on* $[a, b]$. *Then for any* $t \in [a, b]$

$$\int_a^t \alpha'(x)\, dx = \alpha(t) - \alpha(a). \tag{1.3}$$

Remark 1.13 The assumption $\alpha' \in \mathcal{R}$ on $[a, b]$ is needed. Using a construction similar to that of a Cantor set, Volterra constructed a function (not monotone though) which had derivative everywhere in $[0, 1]$ but its derivative is not Riemann integrable. Cantor's function is monotone increasing in $[0, 1]$, equals a constant in any of the middle third interval removed, therefore has derivative equal 0 at any point of those intervals. Since Cantor's set has measure 0, Cantor's function has derivative equal 0 almost everywhere in $[0, 1]$. If we can define an integral for such a function, its integral on any subinterval of $[0, 1]$ must be 0, thus the above equality relation can't hold in such a case.

Remark 1.14 Suppose that $f \in \mathcal{R}(\alpha)$ on $[a,b]$ and $a < c < b$. Then $\int_a^c 1\,d\alpha = \alpha(c) - \alpha(a)$, $\lim_{k\to\infty} \int_a^{c+\frac{1}{k}} 1\,d\alpha = \alpha(c+0) - \alpha(a)$. Since $[a,c] = \cap_k [a, c+\frac{1}{k}]$, or one could think of $\int_a^c 1\,d\alpha$ as $\int \chi_{[a,c]}(x)\,d\alpha$, and since $\lim_{k\to\infty} \chi_{[a,c+\frac{1}{k}]}(x) = \chi_{[a,c]}(x)$, one expects $\lim_{k\to\infty} \int_a^{c+\frac{1}{k}} 1\,d\alpha = \int_a^c 1\,d\alpha$. This would require $\alpha(c+0) = \alpha(c)$. For this reason, one often chooses to work with an α which is right-continuous.

Since $\lim_{k\to\infty} \int_a^{c-\frac{1}{k}} 1\,d\alpha = \alpha(c-0) - \alpha(a)$ and $[a,c) = \cup_k [a, c-\frac{1}{k}]$, it is reasonable to treat $\lim_{k\to\infty} \int_a^{c-\frac{1}{k}} 1\,d\alpha$ as $\int_{[a,c)} 1\,d\alpha$ and $\lim_{k\to\infty} \int_a^{c+\frac{1}{k}} 1\,d\alpha$ as $\int_{[a,c]} 1\,d\alpha$; namely, the notation of integration over a set is more precise than that of integration over a lower and upper limit.

2 The Completion of the Space of Continuous Functions under Integral Norms

We sketch below some main properties of the Completion of the Space of Continuous Functions under Integral Norms. The main line of approach is adapted from the following article by P. Lax, *Rethinking the Lebesgue Integral*, The American Mathematical Monthly, Dec., 2009, Vol. 116, No. 10 (Dec., 2009), pp. 863-881. Although the discussion can be done with respect to the Riemann-Stieltjes integral, we limit our discussion to the standard Riemann integral.

For any $1 \le p < \infty$, let $L^p[a,b]$ denote the completion of $C[a,b]$ under the norm $\|f\|_{L^p[a,b]} := \left(\int_a^b |f(x)|^p\,dx\right)^{1/p}$. From the abstract completion process, each element $\phi \in L^p[a,b]$ is an equivalence class of Cauchy sequences $\{c_k\}$ in $C[a,b]$ under the $L^p[a,b]$-norm. We would like to address the following questions

a. Does each equivalence class of Cauchy sequences $\{c_k\}$ in $C[a,b]$ under the $L^p[a,b]$-norm associate to a pointwise-defined function on $[a,b]$?

b. Does each equivalence class of Cauchy sequences $\{c_k\}$ in $C[a,b]$ under the $L^p[a,b]$-norm associate to a well-defined integral? What usual properties of integrals are preserved?

c. Establish criteria for other limiting process of functions in $C[a,b]$ or $L^p[a,b]$, e.g., pointwise limit, that result in a limit in $L^p[a,b]$.

An elementary observation. Suppose that $\{c_k\}$ is a Cauchy sequence in $C[a,b]$ under the $L^p[a,b]$-norm. Then for $[c,d] \subset [a,b]$, $\int_c^d c_k(x)\,dx$ is a Cauchy sequence in $\mathbb{R}$; and for any $1 \le q \le p$, $\|c_k\|_{L^q[c,d]}$ is a Cauchy sequence in $\mathbb{R}$. Furthermore, $\lim_{k\to\infty} \int_c^d c_k(x)\,dx$ and $\lim_{k\to\infty} \|c_k\|_{L^q[c,d]}$ remain the same if $\{c_k\}$ is replaced by an equivalent Cauchy sequence.

These properties are simple consequences of the triangle inequalities:

$$
\begin{aligned}
\left|\int_c^d c_k(x)\,dx - \int_c^d c_l(x)\,dx\right| &\le \int_c^d |c_k(x) - c_l(x)|\,dx \\
&\le \left(\int_c^d |c_k(x) - c_l(x)|^p\,dx\right)^{1/p} (d-c)^{1-1/p}, \\
\left|\|c_k\|_{L^q[c,d]} - \|c_l\|_{L^q[c,d]}\right| &\le \|c_k - c_l\|_{L^q[c,d]} \\
&\le \|c_k - c_l\|_{L^p[c,d]} (d-c)^{1/q-1/p}.
\end{aligned}
$$

As a result, if we denote by ϕ the equivalence class of $\{c_k\}$, it makes sense to define

$$\int_c^d \phi = \lim_{k\to\infty} \int_c^d c_k(x)\,dx, \quad \|\phi\|_{L^q[c,d]} = \lim_{k\to\infty} \|c_k\|_{L^q[c,d]}.$$

Any $c \in C[a,b]$ defines a Cauchy sequence $\{c_k = c\}$ in $L^p[a,b]$ and $\|c\|_{L^q[c,d]}$ as defined through the limiting process above equals the definition through Riemann's integral.

It also follows routinely that if $\{[c_i, d_i]\}$ is a finite collection of disjoint subintervals of $[a,b]$, then

$$\int_{\cup_i [c_i,d_i]} \phi = \sum_i \int_{[c_i,d_i]} \phi$$

is well defined; the intervals $\{[c_i, d_i]\}$ can also be replaced by open or half-open intervals. Furthermore, if $\phi, \psi \in L^p[a,b]$, then for any scalars a, b,

$$\int_{[a,b]} (a\phi + b\psi) = a\int_{[a,b]} \phi + b\int_{[a,b]} \psi.$$

Any open set G of $\mathbb{R}$ is a union of at most countably many open intervals (c_i, d_i). For any $c \in C[a,b]$ and any open subset G of $[a,b]$, the integrals $\int_G c(x)\,dx$ and $\int_G |c(x)|^p\,dx$ have clearly defined meanings as the (absolutely convergent) sum of the integrals on each constitutive open intervals, with $\int_G 1\,dx = |G|$, the length of G .

Definition 2.1 Uniform Absolute Continuity of A Sequence of Integrals. The integrals of a sequence of functions $\{c_k\}$ in $C[a,b]$ are said to be **uniformly absolutely continuous** on $[a,b]$ if for any $\epsilon > 0$, there exists a $\delta > 0$ such that for any open subset G of $[a,b]$ with its length $|G| < \delta$,

$$\int_G |c_k(x)|^p\,dx < \epsilon \text{ for all } k.$$

$\Diamond$

Proposition 2.2 Uniform Absolute Continuity of the Integrals of A Cauchy Sequence in $C[a,b]$. *The integrals of any Cauchy sequence $\{c_k\}$ in $C[a,b]$ under the $L^p[a,b]$-norm are uniformly absolutely continuous on $[a,b]$.*

Proof. For the given $\epsilon > 0$, there exists some N such that for all $k, l \geq N$, $\|c_k - c_l\|_{L^p[a,b]} < \epsilon^{1/p}/2$. Since $c_i \in C[a,b]$ for $i = 1, \cdots, N$, there exists a common $M > 0$ such that $|c_k(x)| \leq M$ for all $x \in [a,b]$ and $1 \leq k \leq N$, therefore, there exists $\delta > 0$ such that for any open subset G of $[a,b]$ with its length $|G| < \delta$, we have

$$\int_G |c_k(x)|^p\,dx < \epsilon/2^p \text{ for } 1 \leq k \leq N.$$

Now for $l > N$, the triangle inequality continues to hold for integrals on G, which implies that

$$\|c_l\|_{L^p(G)} \leq \|c_l - c_N\|_{L^p(G)} + \|c_N\|_{L^p(G)} \leq \epsilon^{1/p}/2 + \epsilon^{1/p}/2.$$

from which our conclusion follows. ■

Exercise 2.3 In the context of Proposition 2.2, suppose that the open set G is decomposed as the union of at most countably many disjoint open intervals:

$G = \cup_l I_l$, and set $a_{k,l} = \int_{I_l} |c_k(x)|^p \, dx$. Prove that

$$\lim_{k\to\infty} \sum_l a_{k,l} = \sum_l \lim_{k\to\infty} a_{k,l}.$$

This allows to define $\int_G |\phi(x)|^p \, dx$ as $\sum_l \int_{I_l} |\phi(x)|^p \, dx$, given by $\lim_{k\to\infty} \int_G |c_k(x)|^p \, dx$.

As a consequence, prove the **absolute continuity of the integral** of $|\phi|^p$, namely, for any $\phi \in L^p[a,b]$ and for any $\epsilon > 0$, there exists a $\delta > 0$ such that for any open subset G of $[a,b]$ with its length $|G| < \delta$,

$$\int_G |\phi(x)|^p \, dx < \epsilon.$$

Exercise 2.4 Construct $b_{k,l} \geq 0$ for k, l such that $\lim_{k\to\infty} b_{k,l}$ exists for each l and

$$\lim_{k\to\infty} \sum_l b_{k,l} \neq \sum_l \lim_{k\to\infty} b_{k,l}.$$

Exercise 2.5 Construct a sequence of non-negative functions $\{c_k(x)\}$ in $C[0,1]$ such that $\lim_{k\to\infty} c_k(x) = 0$ for each $x \in [0,1]$ but $\lim_{k\to\infty} \int_0^1 c_k(x) \, dx > 0$. Verify that the integrals of this sequence fail to be uniformly absolutely continuous on $[0,1]$.

Definition 2.6 Negligible set. A point set is called negligible if it can be covered by an open set of arbitrarily small volume, namely, for any $\epsilon > 0$, there exists an open cover G such that $|G| < \epsilon$. $\Diamond$

The notion of negligible set here is a special case in the context of Lebesgue measure on $\mathbb{R}$ of the notion of a set of measure 0. Note that the union of at most countable number of negligible sets is still negligible.

When two functions are equal except on a negligible set, we use the customary language that they are equal *a.e.*.

Definition 2.7 Realization of a Cauchy Sequence in $L^p[a,b]$. A function $\phi(x)$ defined *a.e.* on $[a,b]$ is said to be a realization of $\phi \in L^p[a,b]$ if there exists a Cauchy sequence $\{c_k\}$ of continuous functions in $[a,b]$ in the equivalence class of ϕ which converges a.e to $\phi(x)$:

$$\lim_{k\to\infty} c_k(x) \to \phi(x) \ a.e..$$

$\Diamond$

Proposition 2.8 Realization of elements of $L^p[a,b]$.

(i) Any Cauchy sequence $\{c_k\}$ in $C[a,b]$ under the $L^p[a,b]$-norm has a subsequence which converges pointwise a.e. on $[a,b]$. Furthermore, for any $\epsilon > 0$, there exists an open set G with $|G| < \epsilon$ such that $\{c_k\}$ converges uniformly over G^c.

(ii) If $\{c_k\}$ and $\{\tilde{c}_k\}$ are two equivalent Cauchy sequences in $C[a,b]$ under the $L^p[a,b]$-norm such that both $\lim_{k\to\infty} c_k(x)$ and $\lim_{k\to\infty} \tilde{c}_k(x)$ exist a.e., then $\lim_{k\to\infty} c_k(x) = \lim_{k\to\infty} \tilde{c}_k(x)$ a.e..

As a result, any ϕ in $L^p[a,b]$, namely, an equivalence class of Cauchy sequences in $C[a,b]$ under the $L^p[a,b]$-norm, has an a.e. pointwise defined realization defined on $[a,b]$. We will denote such a realization by $\phi(x)$. In particular, if ϕ in $L^p[a,b]$ is such that $\|\phi\|_{L^p[a,b]} = 0$, then $\phi(x) = 0$ a.e. on $[a,b]$.

(iii) If $\{c_k\}$ and $\{\tilde{c}_k\}$ are two Cauchy sequences in $C[a,b]$ under the $L^p[a,b]$-

norm such that $\lim_{k\to\infty} c_k(x) = \lim_{k\to\infty} \tilde{c}_k(x)$ *a.e., then* $\{c_k\}$ *and* $\{\tilde{c}_k\}$ *are equivalent Cauchy sequences in* $L^p[a,b]$.

Proof. Any Cauchy sequence $\{c_k\}$ in $C[a,b]$ under the $L^p[a,b]$-norm has a subsequence, still denoted as $\{c_k\}$, such that

$$\|c_k - c_{k+1}\|_{L^p[a,b]} \le \epsilon_k^2,$$

where $\epsilon_k > 0$ are such that $\sum_k \epsilon_k$ converges.

Define

$$D_k = \{x \in [a,b] : |c_k(x) - c_{k+1}(x)| > \epsilon_k\}.$$

Then D_k is open and it follows from

$$\epsilon_k^p |D_k| \le \int_{[a,b]} |c_k(x) - c_{k+1}(x)|^p \, dx \le \epsilon_k^{2p}$$

that $|D_k| \le \epsilon_k^p$.

Note that $\cup_{k=N}^{\infty} D_k$ is open and

$$|\cup_{k=N}^{\infty} D_k| \le \sum_{k=N}^{\infty} |D_k| \le \sum_{k=N}^{\infty} \epsilon_k^p \to 0,$$

as $N \to \infty$. Therefore, the set E^1 defined by

$$E = \cap_{N=1}^{\infty} \cup_{k=N}^{\infty} D_k$$

is negligible, and for any $x \in E^c$, there exists some N_x such that $x \in \cap_{k=N}^{\infty} D_k^c$, which makes $\{c_k(x)\}$ a Cauchy sequence in $\mathbb{R}$, therefore $\lim_{k\to\infty} c_k(x)$ exists. Furthermore, for any $\epsilon > 0$, there exists some N_ϵ such that $|\cup_{k=N}^{\infty} D_k| < \epsilon$. On the complement of $\cup_{k=N}^{\infty} D_k$, $|c_k(x) - c_{k+1}(x)| \le \epsilon_k$ for all $k \ge N$, implying that $\{c_k(x)\}$ converges uniformly on this set.

For (ii), we can use the equivalent Cauchy sequences $\{c_k\}$ and $\{\tilde{c}_k\}$ to construct a new Cauchy sequence, say, with c_k as the $(2k-1)$-th term and $\{\tilde{c}_k\}$ as the $(2k)$-th term, then appeal to the proof of (i), making sure that in selecting the subsequence, infinitely many terms from both sequences are selected. This subsequence, selected from two *a.e.* convergent sequences, converges *a.e.*, which shows that $\lim_{k\to\infty} c_k(x) = \lim_{k\to\infty} \tilde{c}_k(x)$ *a.e.*.

For (iii), define $f_k(x) = c_k(x) - \tilde{c}_k(x)$, then $\{f_k(x)\}$ is a Cauchy sequence in $C[a,b]$ under the $L^p[a,b]$-norm such that $\lim_{k\to\infty} f_k(x) = 0$ *a.e.*. We now show that $\{f_k(x)\} \to 0$ in $L^p[a,b]$. For any $\epsilon > 0$, first apply Proposition 2.2 to $\{f_k(x)\}$ to find $\delta > 0$ such that for any open set G in $[a,b]$ with $|G| < \delta$, we have $\int_G |f_k(x)|^p \, dx < \epsilon$. Then apply the proof for (i) to $\{f_k(x)\}$ to find an open set G in $[a,b]$ with $|G| < \delta$ such that $f_k \to 0$ uniformly in G^c. Thus there exists N such that $|f_k(x)|^p \le \epsilon$ for all $x \in G^c$ and $k \ge N$. We would like to estimate $\int_{[a,b]} |f_k(x)|^p \, dx$ by $\int_G |f_k(x)|^p \, dx + \int_{G^c} |f_k(x)|^p \, dx$. But in the Riemann integral setting, integration over an arbitrary closed set is not defined. We complete the argument in the following way. Since $|f_k(x)|^p \in \mathcal{R}[a,b]$, we can find a partition $a = a_0 < a_1 < \cdots < a_{N_k} = b$ such that

$$\left| \int_a^b |f_k(x)|^p \, dx - \sum_{i=1}^{N_k} m_i(|f_k|^p)(a_i - a_{i-1}) \right| < \epsilon.$$

For any interval $[a_{i-1}, a_i]$ that has non-empty intersection with G^c, we see that $m_i(|f_k|^p) = \inf_{[a_{i-1},a_i]} |f_k|^p \le \epsilon$; the remaining subintervals $[a_{i-1}, a_i]$ are contained in G, and the lower Riemann sum over such intervals $\le \int_G |f_k(x)|^p \, dx \le \epsilon$ for any $k \ge N$. It then follows that for any $k \ge N$,

$$\int_a^b |f_k(x)|^p \, dx \le (b-a)\epsilon + 2\epsilon.$$

proving that $\int_a^b |f_k(x)|^p \, dx \to 0$ as $k \to \infty$. ■

Remark 2.9 As a consequence of Proposition 2.8, for any realization $\phi(x)$ of some $\phi \in L^p[a,b]$ and any $\epsilon > 0$, there exists a closed set F of $[a,b]$ with $|F^c| < \epsilon$ such that the *restriction* of $\phi(x)$ on F is continuous. Note that this is not saying that $\phi(x)$ is continuous at every point of F as a function on $[a,b]$. Furthermore, for any $\phi \in L^p[a,b]$ and any $\epsilon, \epsilon' > 0$, one can find a continuous approximation $c \in C[a,b]$ in the sense that exists an open set G with $|G| < \epsilon'$ such that

$$\|\phi - c\|_{L^p[a,b]} < \epsilon, \quad \text{and } |\phi(x) - c(x)| < \epsilon \text{ for all } x \in G^c.$$

Here $\|\phi - c\|_{L^p[a,b]}$ is defined through the limiting process instead of the Riemann integral for a general $\phi \in L^p[a,b]$.

Exercise 2.10 Let $f \in L^1[a,b]$. Extend f to be 0 outside of $[a,b]$. Prove that $\int_a^b |f(x+h) - f(x)| \, dx \to 0$ as $|h| \to 0$.

Exercise 2.11 Let $f \in L^1[a,b]$. Prove that $F(t) := \int_a^t f(x) \, dx$ is a continuous function of $x \in [a,b]$.

Exercise 2.12 Let $f \in L^1[a,b]$. Extend f to be 0 outside of $[a,b]$ and define $f_h(x) = h^{-1} \int_x^{x+h} f(y) \, dy$ for $h > 0$ small. Prove that $\int_a^b |f_h(x) - f(x)| \, dx \to 0$ as $h \to 0$.

Corollary 2.13 **$L^p[a,b]$ is closed under taking absolute values and maximums.** *If $\{c_k\}$ in $C[a,b]$ is a Cauchy sequence in $L^p[a,b]$-norm, representing ϕ in $L^p[a,b]$, then $\{|c_k|\}$ in $C[a,b]$ is a Cauchy sequence in $L^p[a,b]$-norm whose realization is given by $|\phi(x)|$.*

If $\phi, \psi \in L^p[a,b]$, then $\max\{\phi, \psi\} := (\phi + \psi + |\phi - \psi|)/2 \in L^p[a,b]$ and $\min\{\phi, \psi\} := (\phi + \psi - |\phi - \psi|)/2 \in L^p[a,b]$.

Proposition 2.14 **Rapidly Convergent Sequence in $L^p[a,b]$.** *Let $\{\phi_k\}$ be a sequence of elements in $L^p[a,b]$ converging rapidly in the sense*

$$\|\phi_k - \phi_{k+1}\|_{L^p[a,b]} < \epsilon_k^2$$

where $\sum_k \epsilon_k$ converges. Then there exists a unique $\phi \in L^p[a,b]$ such that $\|\phi_k - \phi\|_{L^p[a,b]} \to 0$ and $\phi_k(x) \to \phi(x)$ a.e. as $k \to \infty$.

Proof. For each ϕ_k there exists some $c_k \in C[a,b]$ and an open set G_k with $|G_k| < \epsilon_k$ such that

$$\|\phi_k - c_k\|_{L^p[a,b]} < \epsilon_k^2, \text{ and } |\phi_k(x) - c_k(x)| < \epsilon_k^2 \text{ for all } x \in G_k^c.$$

Then the set $E = \cap_{l=1}^\infty \cup_{k=l}^\infty G_k$ is negligible, as $|\cup_{k=l}^\infty G_k| \le \sum_{k=l}^\infty |G_k| \to 0$ as $l \to \infty$, and for any $x \in E^c$, there exists some l_x such that $x \in \cap_{k=l_x}^\infty G_k^c$. This implies that $\{c_k\}$ satisfies

$$\|c_k - c_{k+1}\|_{L^p[a,b]} < 2\epsilon_k^2 + \epsilon_{k+1}^2 \text{ and } |\phi_k(x) - c_k(x)| < \epsilon_k^2 \text{ for } k \ge l_x.$$

Then Proposition 2.8 applied to $\{c_k\}$ implies that there exists a unique $\phi \in L^p[a,b]$ such that

$$\|c_k - \phi\|_{L^p[a,b]} \to 0 \text{ and } c_k(x) \to \phi(x) \text{ a.e. as } k \to \infty.$$

We can then conclude our proof by appealing to the triangle inequalities. ■

[1] E is the set of points that belong to infinitely many D_k's, and E^c is then the set of points that belong to at most a finite number of D_k's.

Definition 2.15 Nonnegative Elements of $L^p[a,b]$. An element $\phi \in L^p[a,b]$ is said to be non-negative if its realization $\phi(x)$ satisfies $\phi(x) \geq 0$ *a.e.* on $[a,b]$. $\Diamond$

Proposition 2.16 Characterization of Nonnegative Elements of $L^p[a,b]$. *An element $\phi \in L^p[a,b]$ is non-negative iff there exists a Cauchy sequence $\{c_k\}$ representing ϕ such that $c_k^- := \min\{c_k(x), 0\}$ satisfies $\|c_k^-\|_{L^p} \to 0$ as $k \to \infty$.*

As a consequence, if $\phi \in L^p[a,b]$ is non-negative, then $\int_{[a,b]} \phi \geq 0$.[2]

Proof. Suppose that $\{c_k\}$ is a sequence in $C[a,b]$, Cauchy in $L^p[a,b]$-norm, representing ϕ such that $\|c_k^-\|_{L^p} \to 0$ as $k \to \infty$. Define $c_k^+ = c_k - c_k^-$, which is identified as $\max\{c_k(x), 0\}$. Then $\{c_k^+\}$ is a sequence in $C[a,b]$, Cauchy in $L^p[a,b]$-norm. Proposition 2.8 applied to both $\{c_k^+\}$ and $\{c_k^-\}$ implies that there exists a subsequence, still indexed by k to avoid extra indices, such that $\lim_{k\to\infty} c_k^+(x)$ exists *a.e.*, which is evidently ≥ 0 *a.e.*, and $\lim_{k\to\infty} c_k^-(x) \to 0$ *a.e.* (refer to (ii) of Proposition 2.8). Therefore

$$\lim_{k\to\infty} c_k(x) = \lim_{k\to\infty} c_k^+(x) + \lim_{k\to\infty} c_k^-(x) \geq 0 \text{ a.e.}$$

Conversely, suppose that $\phi(x) \geq 0$ *a.e.* and $\{c_k\}$ is a sequence in $C[a,b]$, Cauchy under the $L^p[a,b]$ norm representing ϕ and $c_k(x) \to \phi(x)$ *a.e.*. Then we must have $\lim_{k\to\infty} c_k^-(x) \to 0$ *a.e.*. Noting that

$$|c_k^-(x) - c_l^-(x)| \leq |c_k(x) - c_l(x)|$$

we see that $\{c_k^-\}$ is also a sequence in $C[a,b]$, Cauchy under the $L^p[a,b]$ norm. Then the proof of of Proposition 2.8 shows that $c_k^- \to 0$ in $L^p[a,b]$ norm. ■

Theorem 2.17 Monotone Convergence Theorem. *Suppose that $\{\phi_k\}$ is a sequence in $L^1[a,b]$ such that $0 \leq \phi_k(x) \leq \phi_{k+1}(x)$ a.e. for all k and that $\int_a^b \phi_k$ is bounded. Then the a.e. defined $\phi(x) := \lim_{k\to\infty} \phi_k(x)$ (it could take value ∞) is the realization of some $\phi \in L^1[a,b]$, and $\|\phi_k - \phi\|_{L^1[a,b]} \to 0$ as $k \to \infty$.*

Proof. First, Proposition 2.16 implies that $\int_a^b \phi_k(x)\,dx \leq \int_a^b \phi_{k+1}(x)\,dx$, so the sequence of scalars $\{\int_a^b \phi_k(x)\,dx\}$ as a bounded, monotone increasing sequence is convergent. Since for $l > k$, $\int_a^b |\phi_l(x) - \phi_k(x)|\,dx = \int_a^b (\phi_l(x) - \phi_k(x))\,dx$, it follows that $\{\phi_k\}$ is a Cauchy sequence in $L^1[a,b]$. Then by Proposition 2.14 it has a subsequence converging to some $\tilde{\phi}$ in $L^1[a,b]$, and to $\tilde{\phi}(x)$ *a.e.* pointwise. This identifies $\phi(x) := \lim_{k\to\infty} \phi_k(x)$ with $\tilde{\phi}(x)$ *a.e.* pointwise, showing that $\phi \in L^1[a,b]$. Since the full sequence $\{\phi_k\}$ is a Cauchy sequence in $L^1[a,b]$, it follows that $\phi_k \to \phi$ in $L^1[a,b]$. ■

Exercise 2.18 Suppose that $\{\phi_k\}$ is a sequence in $L^1[a,b]$ such that $\phi_k(x) \geq \phi_{k+1}(x)$ *a.e.* for all k and that $\int_a^b \phi_k$ is bounded. Prove that the *a.e.* defined $\phi(x) := \lim_{k\to\infty} \phi_k(x)$ is the realization of some $\phi \in L^1[a,b]$, and $\|\phi_k - \phi\|_{L^1[a,b]} \to 0$ as $k \to \infty$.

Exercise 2.19 Suppose that $\{\phi_k\}$ is a sequence in $L^1[a,b]$ such that $\phi_k(x) \geq 0$ *a.e.* for all k and that $\sum_{k=1}^l \int_a^b \phi_k$ is bounded independent of k. Prove that $\lim_{l\to\infty} \sum_{k=1}^l \phi_k(x)$ is the realization of some of element in $L^1[a,b]$. Denoting

[2] Since $\int_{[a,b]} \phi$ is defined through a limiting process via an approximate sequence not via a Riemann sum using $\phi(x)$, the sign of $\int_{[a,b]} \phi$ does not automatically follow from $\phi(x) \geq 0$ *a.e.*.

it as $\sum_{k=1}^{\infty} \phi_k$, prove that

$$\int_a^b \left(\sum_{k=1}^{\infty} \phi_k \right) dx = \sum_{k=1}^{\infty} \left(\int_a^b \phi_k \, dx \right).$$

Theorem 2.20 Fatou Theorem. *Suppose that $\{\phi_k\}$ is a sequence in $L^1[a,b]$ such that $\phi_k(x) \geq 0$ a.e. for all k and that $\liminf_{k\to\infty} \int_a^b \phi_k(x)$ is finite. Then*

$$\liminf_{k\to\infty} \phi_k(x) := \lim_{l\to\infty} \inf\{\phi_k(x) : k \geq l\} \in L^1[a,b]$$

and

$$\int_a^b \liminf_{k\to\infty} \phi_k(x) \leq \liminf_{k\to\infty} \int_a^b \phi_k(x).$$

Proof. Define $\psi_{k,l} = \min\{\phi_k, \cdots, \phi_{k+l}\}$. Then $\psi_{k,l} \in L^1[a,b]$ and $\psi_{k,l}(x) \geq \psi_{k,l+1}(x)$ *a.e.*. Then Theorem 2.17 implies that $\Phi_k = \lim_{l\to\infty} \psi_{k,l} \in L^1[a,b]$ with $\int_a^b \Phi_k = \lim_{l\to\infty} \int_a^b \psi_{k,l}$. Since for any $l > l' \geq k$, $\psi_{k,l} \leq \phi_{l'}$, it follows that $\int_a^b \Phi_k \leq \int_a^b \phi_{l'}$, therefore, $\int_a^b \Phi_k \leq \inf\{\int_a^b \phi_{l'} : l' \geq k\}$.

Now $\Phi_k \leq \Phi_{k+1}$ and $\int_a^b \Phi_k$ is bounded under our assumption. Then Theorem 2.17 implies $\lim_{k\to\infty} \Phi_k \in L^1[a,b]$ and

$$\int_a^b \liminf_{k\to\infty} \phi_k(x) = \int_a^b \lim_{k\to\infty} \Phi_k = \lim_{k\to\infty} \int_a^b \Phi_k \leq \lim_{k\to\infty} \inf\{\int_a^b \phi_{l'} : l' \geq k\},$$

which equals $\liminf_{k\to\infty} \int_a^b \phi_k(x)$. ■

Exercise 2.21 Construct a sequence $\{\phi_k\}$ in $L^1[a,b]$ such that $\phi_k(x) \geq 0$ *a.e.* for all k, and

$$\int_a^b \liminf_{k\to\infty} \phi_k(x) < \liminf_{k\to\infty} \int_a^b \phi_k(x).$$

Theorem 2.22 Dominated Convergence Theorem. *Suppose that $\{\phi_k\}$ is a sequence in $L^1[a,b]$ such that $\lim_{k\to\infty} \phi_k(x) = \phi(x)$ exists a.e., and there exists a $g \in L^1[a,b]$ such that $|\phi_k(x)| \leq g(x)$ a.e.. Then $\phi \in L^1[a,b]$ and $\phi_k \to \phi$ in $L^1[a,b]$. As a consequence, $\int_a^b \phi_k \to \int_a^b \phi$ as $k \to \infty$.*

Proof. Define $\psi_{k,l} = \min\{\phi_k, \cdots, \phi_{k+l}\}$. Then $\psi_{k,l} \in L^1[a,b]$ and $\psi_{k,l}(x) \geq \psi_{k,l+1}(x)$ *a.e.*. ϕ_k and $\psi_{k,l}$ are no longer necessarily non-negative, but $|\psi_{k,l}(x)| \leq g(x)$ *a.e.*. Then $\{g(x) \pm \psi_{k,l}(x)\}$ are non-negative functions in $L^1[a,b]$, monotone in l. Therefore, $\lim_{l\to\infty}(g(x) \pm \psi_{k,l}(x)) \in L^1[a,b]$, with

$$\int_a^b \left(g \pm \lim_{l\to\infty} \psi_{k,l} \right) = \lim_{l\to\infty} \int_a^b (g \pm \psi_{k,l}),$$

from which it follows that $\Phi_k = \lim_{l\to\infty} \psi_{k,l} \in L^1[a,b]$, with

$$\int_a^b \Phi_k = \lim_{l\to\infty} \int_a^b \psi_{k,l}.$$

Note that it follows from $\phi_k(x) \to \phi(x)$ *a.e.* that $\Phi_k(x) \to \phi(x)$ *a.e.* as $k \to \infty$. Therefore by Theorem 2.17 $\phi \in L^1[a,b]$.

We can now apply Fatou Theorem to the sequence of non-negative functions $2g - |\phi_k - \phi| \in L^1[a,b]$ to imply

$$\int_a^b \liminf_{k\to\infty} (2g - |\phi_k - \phi|) \leq \liminf_{k\to\infty} \int_a^b (2g - |\phi_k - \phi|) = \int_a^b 2g - \limsup_{k\to\infty} \int_a^b |\phi_k - \phi|.$$

Since $\liminf_{k\to\infty}(2g - |\phi_k - \phi|) = 2g$, it follows that $\limsup_{k\to\infty}\int_a^b |\phi_k - \phi| = 0$. ■

Exercise 2.23 Suppose that $\{\phi_k\}$ is a sequence in $L^1[a,b]$ and that $\sum_{k=1}^{\infty}\int_a^b |\phi_k|$ is convergent. Prove that $\lim_{l\to\infty}\sum_{k=1}^{l}\phi_k(x)$ is the realization of some of element in $L^1[a,b]$. Denoting it as $\sum_{k=1}^{\infty}\phi_k$, prove that

$$\int_a^b \left(\sum_{k=1}^{\infty}\phi_k\right) dx = \sum_{k=1}^{\infty}\left(\int_a^b \phi_k\, dx\right).$$

3 The Space of Riemann Integrable Functions

Here we discuss the relation between the space of Riemann integrable functions on $[a,b]$ and $L^1[a,b]$, the completion of $C[a,b]$ under the $L^1[a,b]$ norm.

We first make the following definition.

Definition 3.1 Let f be defined on an interval I. The oscillation of a function f over the set $S \subset I$ is defined to be

$$\sup_S f - \inf_S f = M(f,S) - m(f,S)$$

and is denoted as $\mathrm{osc}(f,S)$.

The oscillation of a function f at a point x is defined to be

$$\lim_{r\searrow 0}\mathrm{osc}(f, I\cap I(x,r)),$$

and is denoted as $\mathrm{osc}(f)(x)$. Here $I(x,r)$ is the open interval of radius r centered at x. ◊

Proposition 3.2 Upper semi-continuity of osc$(f)(x)$. *The function osc$(f)(x)$ is upper semi-continuous. As a consequence, for any real number a, the set $\{x : osc(f)(x) \geq a\}$ is closed.*

Proof. For any real number a, if $x_0 \in \{x : \mathrm{osc}(f)(x) < a\}$, then there exists some $r > 0$ such that $\mathrm{osc}(f, I(x_0,r)) < a$. For any $x \in I(x_0,r)$, we observe that $\mathrm{osc}(f)(x) \leq \mathrm{osc}(f, I(x_0,r)) < a$. Thus $I(x_0,r) \subset \{x : \mathrm{osc}(f)(x) < a\}$, proving that the latter is open. ■

Note that

$$\{x : f \text{ discontinuous at } x\} = \cup_{k\in\mathbb{N}}\{y : \mathrm{osc}(f)(y) \geq \frac{1}{k}\}, \tag{3.1}$$

namely, $\{x : f \text{ discontinuous at } x\}$ is a countable union of of the closed sets $\{y : \mathrm{osc}(f)(y) \geq \frac{1}{k}\}$.

Theorem 3.3 Riemann integrability criterion in terms of the oscillation of the integrand. *A bounded real-valued function f defined on the interval I is Riemann integrable iff*

$$\forall\epsilon > 0, \exists \text{ a partition } \mathcal{P} := \{I_\alpha\} \text{ such that } \sum_\alpha osc(f, I_\alpha)|I_\alpha| < \epsilon. \tag{3.2}$$

In particular, if f is continuous on the closed interval I, then f is Riemann integrable on I.

Proof. Suppose that f is Riemann integrable on I and that $\epsilon > 0$ is given. Then

the Darboux integrability criterion gives us a partition $\mathcal{P} := \{I_\alpha\}$ such that

$$\int_I f - \frac{\epsilon}{2} < L(f, \mathcal{P}) \leq \int_I f \leq U(f, \mathcal{P}) < \int_I f + \frac{\epsilon}{2},$$

which implies that

$$\sum_\alpha \operatorname{osc}(f, I_\alpha)|I_\alpha| = U(f, \mathcal{P}) - L(f, \mathcal{P}) < \epsilon.$$

Suppose that (3.2) holds. Then $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \epsilon$, and

$$0 \leq \overline{\int_I} f - \underline{\int_I} f \leq U(f, \mathcal{P}) - L(f, \mathcal{P}) < \epsilon.$$

Since $\epsilon > 0$ is arbitrary, it follows that $\overline{\int_I} f - \underline{\int_I} f = 0$, and f is Riemann integrable on I.

Finally, suppose that f is continuous on the closed interval I, then it is uniformly continuous on I. For any given $\epsilon > 0$, there exists $\delta > 0$ such that for any partition $\mathcal{P} := \{I_\alpha\}$ of I with $\lambda(P) < \delta$, we have $\operatorname{osc}(f, I_\alpha) < \epsilon/|I|$ for all $I_\alpha \in \mathcal{P}$. It then follows that

$$\sum_\alpha \operatorname{osc}(f, I_\alpha)|I_\alpha| \leq \epsilon,$$

proving the Riemann integrability of f on I. ■

Definition 3.4 A set $S \subset \mathbb{R}$ is said to have content 0, if for any $\epsilon > 0$, there exists a *finite* cover $\{I_i\}$ of S by intervals such that

$$\sum_i |I_i| < \epsilon.$$

◊

We are now ready to formulate the following theorem.

Theorem 3.5 Riemann integrability in terms of the set of discontinuity. *A bounded function f on a bounded closed interval I is Riemann integrable on I iff its set of discontinuity is negligible.*

Proof. We will use (3.1) for the only if part.

Suppose that f is Riemann integrable on I. For each $k \in \mathbb{N}$, we will prove that the closed set $D_k := \{y : \operatorname{osc}(f)(y) \geq \frac{1}{k}\}$ is a set of content 0.

Given any $\epsilon > 0$. There exists a partition $\mathcal{P} = \{I_\alpha\}$ of I such that

$$\sum_\alpha \operatorname{osc}(f, I_\alpha)|I_\alpha| < \frac{\epsilon}{2k}.$$

The intervals in $\mathcal{P}$ are divided into two subgroups: the subgroup $\mathcal{L}_k$ consisting those I_α such that $\operatorname{osc}(f, I_\alpha) \geq \frac{1}{2k}$, and the subgroup $\mathcal{S}_k$ consisting those I_α such that $\operatorname{osc}(f, I_\alpha) < \frac{1}{2k}$. Then it follows from

$$\frac{1}{2k} \sum_{I_\alpha \in \mathcal{L}_k} |I_\alpha| \leq \sum_{I_\alpha \in \mathcal{L}_k} \operatorname{osc}(f, I_\alpha)|I_\alpha| < \frac{\epsilon}{2k}$$

that

$$\sum_{I_\alpha \in \mathcal{L}_k} |I_\alpha| < \epsilon.$$

We now claim that

$$D_k \subset \cup_{I_\alpha \in \mathcal{L}_k} I_\alpha.$$

This will show that D_k is a set of content 0.

If the claim were not true, there would exist some $x \in D_k \setminus \cup_{I_\alpha \in \mathcal{L}_k} I_\alpha$. Thus $x \in \cup_{I_\alpha \in \mathcal{S}_k} I_\alpha$. Since the complement of $\cup_{I_\alpha \in \mathcal{L}_k} I_\alpha$ is open, there exists some interval $I(x,r) \subset \cup_{I_\alpha \in \mathcal{S}_k} I_\alpha$. If $x \in$ interior(I_α) for some $I_\alpha \in \mathcal{S}_k$, it would force $\frac{1}{k} \le \text{osc}(f)(x) \le \text{osc}(f, I_\alpha) < \frac{1}{2k}$, which would be a contradiction. So x can only be on the boundary of one or more $I_\alpha \in \mathcal{S}_k$. We can choose $r > 0$ small enough such that any $y \in I(x,r)$ and x will be in one such common interval. Therefore, $|f(y) - f(x)| < \frac{1}{2k}$. This would lead to $\text{osc}(f)(x) \le \text{osc}(f, I(x,r)) < \frac{1}{k}$, contradicting $x \in D_k$.

For the if part, take any $\epsilon > 0$, then choose k such that $k^{-1} < \epsilon$. The closed set $D_k = \{y : \text{osc}(f)(y) \ge \frac{1}{k}\}$ is a subset of the negligible set D, so can be covered by a union G of a finite number of intervals such that $|G| < \epsilon$. Every point $x \in [a,b] \setminus G$ has an open interval $I(x,r)$ such that $\text{osc}(f, I(x,r)) < k^{-1}$. The closed set $[a,b] \setminus G$ can be covered by the union of a finite number of such intervals and one can find a fine enough partition $\mathcal{P} = \{I_i\}$ of $[a,b]$ such that if any I_i contains a point of D_k^c, then $\text{osc}(f, I_i) < k^{-1}$; and the union of those $I_i \subset D_k$ has their sum of lengths $< \epsilon$. This would give

$$\sum_i \text{osc}(f, I_i)|I_i| \le k^{-1}(b-a) + 2\epsilon \sup_{[a,b]} |f|,$$

which proves that f satisfies the Riemann integrability criterion. ■

Corollary 3.6 Approximation Property of Functions in $\mathcal{R}[a,b]$. *Any Riemann integrable function $f \in \mathcal{R}[a,b]$ lies in $L^p[a,b]$ for any $1 \le p < \infty$, and for any $\epsilon > 0$ there exists a continuous function $c \in C[a,b]$ and an open set G in $[a,b]$ such that $|G| < \epsilon$, $f = c$ on G^c, and $\max_{[a,b]} |c| \le \max_{[a,b]} |f|$.*

Proof. Since the set D of discontinuity of f is negligible, for any $\epsilon > 0$ there exists an open set G in $[a,b]$ such that it covers D and $|G| < \epsilon$. For every point x in the closed set $[a,b] \setminus G$, $\text{osc}(f)(x) = 0$, so f is a continuous function on this closed set. It can be extended to a continuous function c on $[a,b]$ such that $\max_{[a,b]} |c| \le \max_{[a,b]} |f|$. Then this c satisfies our requirements. ■

Exercise 3.7 Suppose that $c(x) \in C[a,b]$ and $E \subset [a,b]$ is negligible. Is it true that any modification of $c(x)$ on E into a bounded function $\tilde{c}(x)$ will remain to have a negligible set of discontinuity and remain Riemann integrable on $[a,b]$?

Exercise 3.8 Suppose that E is a closed negligible subset of $[a,b]$. Prove that the set of discontinuity of the characteristic function $\chi_E(x)$ is precisely E.

Exercise 3.9 Suppose that E is the union of at most countably many closed negligible subsets of $[a,b]$. Prove that there is a bounded function $f(x)$ defined on $[a,b]$ whose set of discontinuity is precisely E.

4 Extension of the Fundamental Theorem of Calculus; Absolutely Continuous Functions

Corollary 1.12 extends the Fundamental Theorem of Calculus to those monotone functions α such that α' exists and is Riemann integrable. Here we discuss its further generalization and related issues. Questions that need to be addressed include

(I). Identify conditions that guarantee that α' exists and is Riemann integrable, or *a.e.* exists and is in $L^1[a,b]$.

(II). Does (1.3) hold for any function α such that α' *a.e.* exists and is in $L^1[a,b]$? More generally, under what conditions on a function f on $[a,b]$ there would exist a function $g \in L^1[a,b]$ such that

$$f(t) - f(a) = \int_a^t g(x)\,dx \tag{4.1}$$

holds?

The last question has two parts: (a). The necessary conditions for (4.1) to hold, in particular, does it hold that $f'(x) = g(x)$ for $a.e. x \in [a,b]$? (b). The sufficient conditions for (4.1) to hold.

Definition 4.1 Absolute Continuous Function. A function f defined on $[a,b]$ is said to be **absolutely continuous** on $[a,b]$ if for any $\epsilon > 0$, there exists a $\delta > 0$ such that for any finite collection of disjoint intervals $\{[a_i, b_i] : 1 \le i \le k\}$ of $[a,b]$ with its total length $\sum_{i=1}^k (b_i - a_i) < \delta$,

$$\sum_{i=1}^k |f(b_i) - f(a_i)| < \epsilon. \tag{4.2}$$

$\Diamond$

By Exercise 2.3, if (4.1) holds, then f is absolutely continuous on $[a,b]$.

If f is **Lipschitz continuous** on $[a,b]$, namely, there exists some $L > 0$ such that $|f(x) - f(y)| \le L|y - x|$ for any $x, y \in [a,b]$, then f is absolutely continuous on $[a,b]$.

If $f \in C[a,b]$ is differentiable everywhere in (a,b) and f' is bounded on (a,b), then f is Lipschitz continuous on $[a,b]$, therefore, is absolutely continuous on $[a,b]$.

Cantor's function is differentiable *a.e.* in (a,b), yet is not absolutely continuous on $[a,b]$.

Exercise 4.2 Suppose that $f \in C[a,b]$ is differentiable *a.e.* in (a,b) and $|f'| \le M < \infty$ *a.e.* on (a,b). Is f necessarily Lipschitz continuous on $[a,b]$? Is it necessarily absolutely continuous on $[a,b]$?

We will prove below that any absolutely continuous function on $[a,b]$ is differentiable *a.e.* on $[a,b]$ with its derivative a function in $L^1[a,b]$. Assuming this for now, we prove

Theorem 4.3 Differentiability of an Indefinite Integral. *Suppose that $g \in L^1[a,b]$. Then the function $f(x) := \int_a^x g(t)\,dt$ is differentiable a.e. on $[a,b]$ and $f'(x) = g(x)$a.e. on $[a,b]$.*

Proof. According to our discussion above, $f(x) := \int_a^x g(t)\,dt$ is differentiable *a.e.* on $[a,b]$ and $f'(x) \in L^1[a,b]$. This implies that $g_h(x) := h^{-1}\int_x^{x+h} g(t)\,dt = h^{-1}(f(x+h) - f(x)) \to f'(x)$ *a.e.* on $[a,b]$ as $|h| \to 0$.

On the other hand, according to Exercise 2.12, $\int_a^b |g_h(x) - g(x)|\,dx \to 0$ as $h \to 0$. Then by (i) of Proposition 2.8, there exists a sequence $h_k \to 0$ such that $g_{h_k}(x) \to g(x)$ *a.e.* on $[a,b]$ as $k \to \infty$. This then identifies $f'(x) = g(x)$ *a.e.* on $[a,b]$.

We could also complete the last part by using the Fatou Theorem:

$$\begin{aligned}\int_a^b |f'(x) - g(x)|\,dx &= \int_a^b \lim_{k\to\infty} |g_{h_k}(x) - g(x)|\,dx \\ &\le \liminf k \to \infty \int_a^b |g_{h_k}(x) - g(x)|\,dx = 0,\end{aligned}$$

from which we conclude that $f'(x) = g(x)$ *a.e.* on $[a, b]$ via (ii) of Proposition 2.8. ■

For the remaining part, we will state and use Lebesgue's differentiability theorem of a monotone function but will skip its proof.

Theorem 4.4 Lebesgue's Differentiability Theorem of a Monotone Function. *Any monotone function on $[a, b]$ is differentiable a.e. on $[a, b]$.*

Proposition 4.5 Integrability of the Derivative of a Monotone Function. *Suppose that f is a monotone increasing function on $[a, b]$, then its derivative, $f'(x)$, is the realization of a function in $L^1[a, b]$. Furthermore,*

$$\int_a^b f'(x)\,dx \le f(b) - f(a).$$

Proof. According to Theorem 4.4, $h^{-1}(f(x+h) - f(x)) \to f'(x)$*a.e.* on $[a, b]$. For each $h > 0$, $h^{-1}(f(x+h) - f(x))$ is a non-negative function in $\mathcal{R}[a, b]$ (extend $f(x) = f(b)$ for $x > b$), and

$$\int_a^b h^{-1}(f(x+h)-f(x))\,dx = h^{-1}\left(\int_b^{b+h} f(x)\,dx - \int_a^{a+h} f(x)\,dx\right) \le f(b)-f(a),$$

using $\int_a^{a+h} f(x)\,dx \ge f(a)h$. Thus by Fatou Theorem,

$$\begin{aligned}\int_a^b f'(x)\,dx &= \int_a^b \liminf_{h\to 0+} h^{-1}(f(x+h) - f(x))\,dx \\ &\le \liminf_{h\to 0+} \int_a^b h^{-1}(f(x+h) - f(x))\,dx \le f(b) - f(a).\end{aligned}$$

■

Definition 4.6 Function of Bounded Variation. A function f on $[a, b]$ is said to have bounded variation on $[a, b]$, if there exists some $M > 0$ such that for any partition $\mathcal{P} = \{a = a_0 < a_1 < \cdots < a_k = b\}$,

$$\sum_{i=1}^k |f(a_i) - f(a_{i-1})| \le M.$$

$\sup_{\mathcal{P}} \sum_{i=1}^k |f(a_i) - f(a_{i-1})|$ is called the total variation of f on $[a, b]$ and is denoted as $V(f, [a, b])$. ◊

If f is a monotone increasing function on $[a, b]$, then it has bounded variation on $[a, b]$ and $V(f, [a, b]) = f(b) - f(a)$.

If $f = g - h$ is the difference of two monotone increasing functions on $[a, b]$, then it has bounded variation on $[a, b]$ and $V(f, [a, b]) \le V(g, [a, b]) + V(h, [a, b])$.

Proposition 4.7 *Any function of bounded variation on $[a, b]$ is the difference of two monotone increasing functions.*

Proof. Let f be a function of bounded variation on $[a, b]$. Define $g(x) := V(f, [a, x])$. Then $g(x)$ is a monotone increasing function on $[a, b]$. We now prove that $h(x) := g(x) - f(x)$ is a monotone increasing function on $[a, b]$.

Take $x < y$ in $[a, b]$, then

$$h(y) - h(x) = g(y) - g(x) - [f(y) - f(x)] = V(f, [x, y]) - [f(y) - f(x)].$$

$V(f, [x, y])$ is the supremum of $\sum_{i=1}^k |f(a_i) - f(a_{i-1})|$ for any partition $\{x = a_0 < a_1 < \cdots < a_k = y\}$, so $V(f, [x, y]) \ge |f(y) - f(x)|$. ■

An absolutely continuous function on $[a, b]$ has bounded variation on $[a, b]$, therefore, according to Proposition 4.7, Theorem 4.4 and Proposition 4.5, has its derivative in $L^1[a, b]$.

Theorem 4.8 Fundamental Theorem of Calculus. *A function f on $[a, b]$ satisfies (4.1) for some function $g \in L^1[a, b]$ iff f is absolutely continuous on $[a, b]$. When (4.1) holds, $f'(x) = g(x)$a.e. on $[a, b]$.*

Proof. We already discussed the necessary part. For the sufficient part, suppose that f is absolutely continuous on $[a, b]$. Then, according to our discussion above, $f'(x)$ exists *a.e.* on $[a, b]$ and is an element in $L^1[a, b]$. Set $F(x) = \int_a^x f'(t)\, dt$. Then, according to Theorem 4.3, $F(x)$ is is absolutely continuous on $[a, b]$ and $F'(x) = f'(x)$ *a.e.* on $[a, b]$.

Now $f(x) - F(x)$ is absolutely continuous on $[a, b]$ and $(f(x) - F(x))' = 0$ *a.e.* on $[a, b]$. We can draw our conclusion based on the following Lemma. ■

Lemma 4.9 Absolute Function with Vanishing Derivative. *Suppose that $g(x)$ is absolutely continuous on $[a, b]$ and $g'(x) = 0$a.e. on $[a, b]$. Then g must be a constant function on $[a, b]$.*

We will use the concept and properties of Vitali covering in proving this.

Definition 4.10 Vitali Covering. Suppose that $\Gamma = \{I_\alpha\}$ is a family of intervals covering a set $E \subset (a, b)$. Suppose that for any $\epsilon > 0$ and any $x \in E$, there exists an interval $I \in \Gamma$ such that $x \in I$ and $|I| < \epsilon$. Then Γ is said for form a Vitali covering of E ◊

Lemma 4.11 Vitali Covering Lemma. *Suppose that $\Gamma = \{I_\alpha\}$ is a Vitali covering of $E \subset (a, b)$. Then for any $\epsilon > 0$, there exists a finite collection of disjoint intervals $\{I_i, 1 \le i \le k\}$ of Γ such that $E \setminus \cup_{i=1}^k I_i$ can be covered by an open set G with $|G| < \epsilon$.*

Proof. We may take the I_α to be closed intervals in (a, b). We choose I_i inductively. Choose I_1 from Γ such that $2|I_1| > \sup |I_\alpha|$. After $\{I_i, 1 \le i \le l\}$ are chosen, if $E \setminus \cup_{i=1}^l I_i \neq \emptyset$, define $\delta_{l+1} = \sup\{|I_\alpha| : I_\alpha \cap I_i = \emptyset, i = 1, \cdots, l\}$. Then $\delta_{l+1} > 0$ and we choose some $I_{l+1} \in \Gamma$ such that $I_{l+1} \cap I_i = \emptyset, i = 1, \cdots, l$ and $2|I_{l+1}| > \delta_{l+1}$.

Since this collection of disjoint intervals are all contained in (a, b), $\sum_l |I_l| < \infty$. Thus, for a given $\epsilon > 0$, there exists k such that $\sum_{l>k} |I_l| < \epsilon/5$.

Any $x \in E \setminus \cup_{i=1}^l I_i$ must be contained in some $I \in \Gamma$ such that $I \cap I_i = \emptyset, i = 1, \cdots, k$. Since $|I| > 0$ and $\delta_l \to 0$ as $l \to \infty$, we claim that $I \cap I_l \neq \emptyset$ for some $l > k$, for, whenever $\delta_{l'} < |I|$, $I \cap I_l \neq \emptyset$ for some $l < l'$. Let I_l' be the interval with the same center point as I_l but $|I_l'| = 5|I_l|$, then $I \subset I_l'$ whenever $I \cap I_l \neq \emptyset$. Thus $\cup_{l>k} I_l'$ covers $E \setminus \cup_{i=1}^l I_i$ with $\sum_{l>k} |I_l'| = 5 \sum_{l>k} |I_l| < \epsilon$. ■

Proof of Lemma 4.9. Suppose g is not a constant function on $[a, b]$. Then there exists some $c \in (a, b)$ such that $|g(c) - g(a)| > 2\epsilon$ for some $\epsilon > 0$. Consider $E_c := \{x \in (a, c) : g'(x) = 0\}$. Then $[a, c] \setminus E_c$ is negligible. For any $x \in E_c$ and any $r > 0$, for all sufficiently small $h > 0$, we have $[x, x + h] \subset (a, c)$ and

$$|g(x + h) - g(x)| \le rh.$$

Thus for any fixed such $r > 0$, the family $\{[x, x + h] : x \in E_c\}$ forms a Vitali cover of E_c. Thus, for any $\delta > 0$, there exists disjoint intervals

$$[x_1, x_1 + h_1], \cdots, [x_k, x_k + h_k]$$

such that

$$[a, c] \setminus \cup_{i=1}^k [x_i, x_i + h_i] \subset ([a, c] \setminus E_c) \cup \left(E_c \setminus \cup_{i=1}^k [x_i, x_i + h_i]\right)$$

can be covered by an open set G with $|G| < \delta$.

We may assume that

$$a < x_1 < x_1 + h_1 < x_2 < x_2 + h_2 < \cdots < x_k < x_k + h_k < c.$$

These points, together with a and c, forms a partition of $[a, c]$. The sum of the lengths of those intervals of this partition not-overlapping with any of $(x_i, x_i + h_i)$ is $< \delta$. Then

$$2\epsilon < |g(c) - g(a)| \leq \sum_{i=1}^{k} |g(x_i) - g(x_i + h_i)| + \sum_{i=0}^{k} |g(x_i + h_i) - g(x_{i+1})|$$

where we set $x_0 + h_0 = a$ and $x_{k+1} = c$.

We fix $r > 0$ such that $r(b - a) < \epsilon$. Using the absolute continuity of f on $[a, b]$, there exists some $\delta > 0$ such that for any collection of disjoint intervals $[a_j, b_j]$ with $\sum_j (b_j - a_j) < \delta$, we have $\sum_j |g(b_j) - g(a_j)| < \epsilon$. For this reason, $\sum_{i=0}^{k} |g(x_i + h_i) - g(x_{i+1})| < \epsilon$. But

$$\sum_{i=1}^{k} |g(x_i) - g(x_i + h_i)| \leq \sum_{i=1}^{k} r h_i \leq r(b - a) < \epsilon.$$

We now see a contradiction and must conclude that g is a constant function on $[a, b]$ ■

Exercise 4.12 Suppose that f is absolutely continuous on (a, b) and $|f'| \leq M < \infty$ *a.e.* on (a, b). Is f necessarily Lipschitz continuous on $[a, b]$?

Exercise 4.13 Suppose that f has bounded variation on $[0, 1]$ and is absolutely continuous on $[\epsilon, 1]$ for any $0 < \epsilon < 1$. Furthermore, suppose that $f(x)$ is continuous at $x = 0$. Prove that f is absolutely continuous on $[0, 1]$.